

QuestionsLab2.1.pdf



abcd__1234



Paralelismo



3º Grado en Ingeniería Informática



Facultad de Informática de Barcelona (Fib)
Universidad Politécnica de Catalunya



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 **Gemini 3**

Nuestro modelo más preciso

Sintetiza horas de
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Apuntes Tema 3

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Razonamiento

Estudia con Gemini

Pregunta 1

Completa

Puntuació 1,00
sobre 1,00

[Marca la pregunta](#)

Assume we run the 1.hello program in boada as indicated in each answer. Select the correct answers (one or more):

- ☐ a. The number of Hello Word! messages printed are one when running ./1.hello in the interactive node. OpenMP parallel uses only one thread because we can only use one physical core.
- ☒ b. The number of Hello Word! messages printed are two when running ./1.hello in the interactive node. OpenMP parallel uses two threads because we can use one physical core, with two logical cores.
- ☐ c. The number of Hello Word! messages printed are not deterministic when running ./1.hello in the interactive node. OpenMP parallel may take any number of threads depending on the workload of the system.
- ☒ d. The number of Hello Word! messages printed are 4 when running OMP_NUM_THREADS=4 ./1.hello in the interactive node.

Pregunta 2

Completa

Puntuació 1,00
sobre 1,00

 Marca la
pregunta

Assume we have run the 2.hello program in boada following the instructions in the source code. Select the correct answers (one or more):

- ☐ a. There is a data race condition problem with variable id because it is firstprivate in the parallel region.
- ☒ b. There is a data race condition problem with variable id because it is shared in the parallel region. We can solve it using the private(id) clause in the parallel construct.
- ☒ c. Moving the declaration of variable id inside the parallel region will solve the concurrency problem. In this case, variable id will be private to each implicit task.
- ☒ d. Messages are interleaved because there are threads that may be running in parallel.

Pregunta 3

Completa

Puntuació 1,00
sobre 1,00

 Marca la
pregunta

Assume we have run the 3.how_many program in boada following the instructions in the source code. Select the correct answers (one or more):

- ☐ a. The number of threads returned by `omp_get_num_threads` outside a parallel region is equal to the value set by the last call to `omp_set_num_threads`, if any; otherwise the number set in the `OMP_NUM_THREADS` environment variable.
- ☒ b. Outside any parallel region there is only one thread running.
- ☒ c. The number of threads in a parallel region is defined by the `OMP_NUM_THREADS` environment variable and can be superseded by either calling the `omp_set_num_threads` intrinsic function or by using the `num_threads` clause of the parallel construct.
- ☒ d. The `num_threads` clause of the parallel construct only affects the current parallel region, superseding any value set in previous calls to `omp_set_num_threads`.
- ☐ e. The number of threads in a parallel region only depends on the number of cores you have in the node.

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Pregunta 4

Completa

Puntuació 1,00
sobre 1,00

 [Marca la
pregunta](#)

Assume we have run the 4.data_sharing program in boada following the instructions in the source code. Select the correct answers (one or more):

- ☐ a. Code in the first parallel region has a concurrency problem when updating the shared x variable. This is because sleep(1) makes threads sleep for 1 second.
- ☒ b. Code in the second parallel region solves the concurrency problem but is not correct if we want to increase variable x with the thread number of all the threads participating in the parallel region.
- ☐ c. Code in the third parallel region solves the concurrency problem and is correct because it copies the value of x inside the parallel region.
- ☒ d. Code in the fourth parallel region makes use of the reduction clause reduction(+:x), creating a copy of variable x for each thread and performing a reduction using "+" at the end of the parallel region. The code of the fourth parallel region solves the concurrency problem and is functional.

WUOLAH

Pregunta 5

Completa

Puntuació 1,00
sobre 1,00

 [Marca la pregunta](#)

Assume we have run the 5.datarace program in boada following the instructions in the source code. Select the correct answers (one or more):

- ☐ a. Updating maxvalue is causing a data race condition that can be solved by adding atomic just before updating it.
- ☐ b. Updating maxvalue is causing a data race condition that can be solved by adding critical just before updating it.
- ☒ c. Updating maxvalue is causing a data race condition that can be solved by adding critical just before the if statement. However, this will serialize the execution of the loop iterations.
- ☒ d. Adding clause reduction(max:maxvalue) in the parallel construct helps to both solve the data race condition and maintain the potential parallelism available in the loop.
- ☒ e. Assuming N multiple of the number of threads, we can distribute the iterations to implicit tasks using the following for statement: `for(i=id*N/howmany; i<(id+1)*N/howmany; i++) { // same code ... }`.



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Pregunta 6

Completa

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 Marca la
pregunta

Assume we have run the 6.datarace program in boada, compiled to generate an OpenMP program. Select the correct answers (one or more):

- ☒ a. Updating countmax is causing a data race condition that can be solved by adding atomic just before updating it.
- ☐ b. Updating countmax is causing a data race condition that cannot be solved by adding critical just before updating it.
- ☒ c. Updating countmax is causing a data race condition that can be solved by adding critical just before the if statement. On the other hand, this will serialize the execution of the loop iterations.
- ☒ d. Adding clause reduction(+:countmax) in the parallel construct solves the data race condition.

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Completa

Puntuació 1,00
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Synchronization Overheads: criticals, atomics and reductions. Indicate which of the following sentences is true (one or more):

- ☐ a. The synchronization overheads of critical and atomic clauses do not depend on the number of threads trying to synchronize each other.
- ☒ b. The synchronization overheads of critical is larger than the synchronization overheads of atomic.
- ☐ c. For a large number of threads it is expected that the reduction clause to be much faster than the critical clause because it uses atomic to update the sum variable in each iteration.
- ☒ d. The performance of pi_omp_sumlocal and pi_omp_reduction are much better than the other programs for more than one thread. This is because those programs only have as many synchronizations as the number of threads. On the other hand, atomic and critical versions have a synchronization per loop iteration.

Más info.:
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